

# Elastic Container Service

Elastic Container Service



## **Introduction to ECS**

# *Elastic Container Service (ECS)*



fully-managed **container** orchestration service.  
Highly secure, reliable, and scalable way to run containers



# Components of ECS

## Cluster

Multiple EC2 instances which will house the docker containers.

## Task Definition

A JSON file that defines the configuration of (upto 10) containers you want to run

## Task

Launches containers defined in Task Definition.

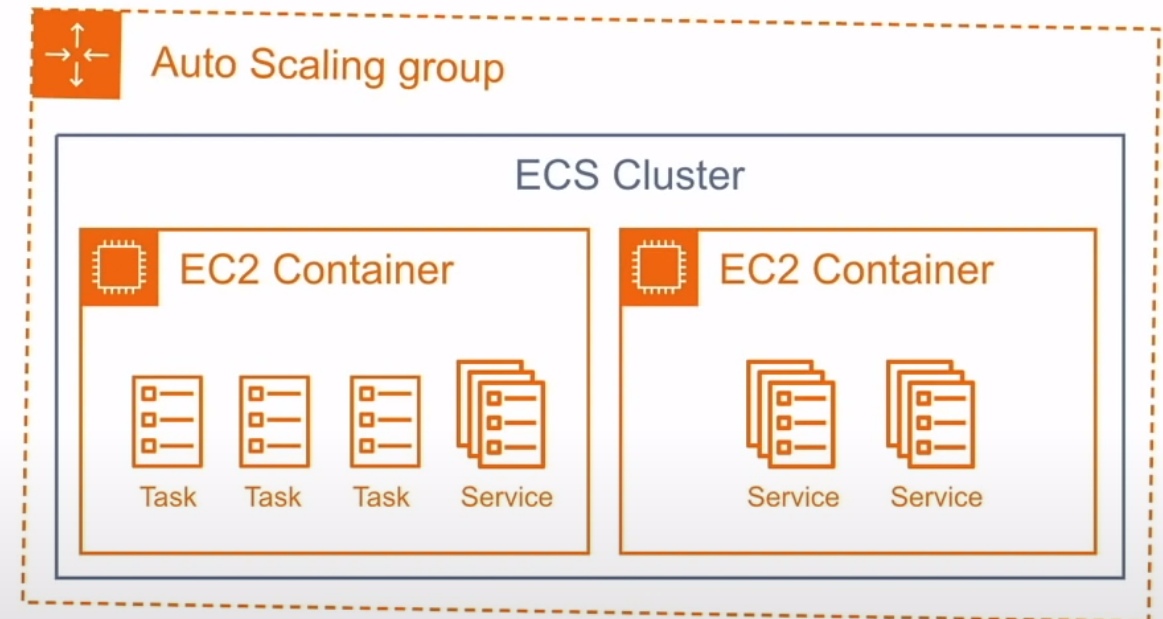
Tasks do not remaining running once workload is complete.

## Service

Ensures tasks remaining running eg. web-app.

## Container Agent

Binary on each EC2 instance which monitors, starts and stops tasks.





# **AWS Developer Associate**

Elastic Container Service



## **Creating a Cluster**





# Creating an ECS Cluster

## Create Cluster

- Use Spot or On Demand
- EC2 Instance Type
- Number of Instances
- EBS Storage Volume
- EC2 can be **Amazon Linux 2** or **Amazon Linux 1**
- Choose a VPC or create a new VPC
- Assign an IAM Role
- Option to turn on CloudWatch Container Insights
- \*Choose a Key Pair

*\*You can SSH into an EC2 Container Instance and make changes but its not generally recommended.*

## EC2 Linux + Networking

### Resources to be created:

Cluster

VPC

Subnets

Auto Scaling group with Linux AMI

EC2 instance type\*

t3a.large

☐ Manually enter desired instance type

Number of instances\*

1

EC2 Ami Id\*

Amazon Linux 2 AMI [ami-084f...

Container instance IAM role

ecsInstanceRole

CloudWatch Container Insights

☐

Enable Container Insights

Key pair

None - unable to SSH

You will not be able to SSH into your EC2 instances without a key pair. You can create a new key pair in the [EC2 console](#).

VPC

Create a new VPC

CIDR block

10.0.0.0/16

Subnet 1

10.0.0.0/24

Subnet 2

10.0.1.0/24

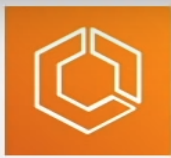


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## Task Definitions File



# Task Definition JSON File Example

## Create new Task Definition

You can define multiple containers within a task definition.

The (docker) images can be provided either via **ECR** or an official docker repository eg. Docker Hub

You must have one **essential** container. If this container fails or stops than all other containers will be stopped.

AWS had a wizard to create Task Definitions instead having to create a file by hand

```
{
  "containerDefinitions": [
    {
      "name": "wordpress",
      "links": [
        "mysql"
      ],
      "image": "wordpress",
      "essential": true,
      "portMappings": [
        {
          "containerPort": 80,
          "hostPort": 80
        }
      ],
      "memory": 500,
      "cpu": 10
    },
    {
      "environment": [
        {
          "name": "MYSQL_ROOT_PASSWORD",
          "value": "password"
        }
      ],
      "name": "mysql",
      "image": "mysql",
      "cpu": 10,
      "memory": 500,
      "essential": true
    }
  ],
  "family": "hello_world"
}
```



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Elastic Container Service

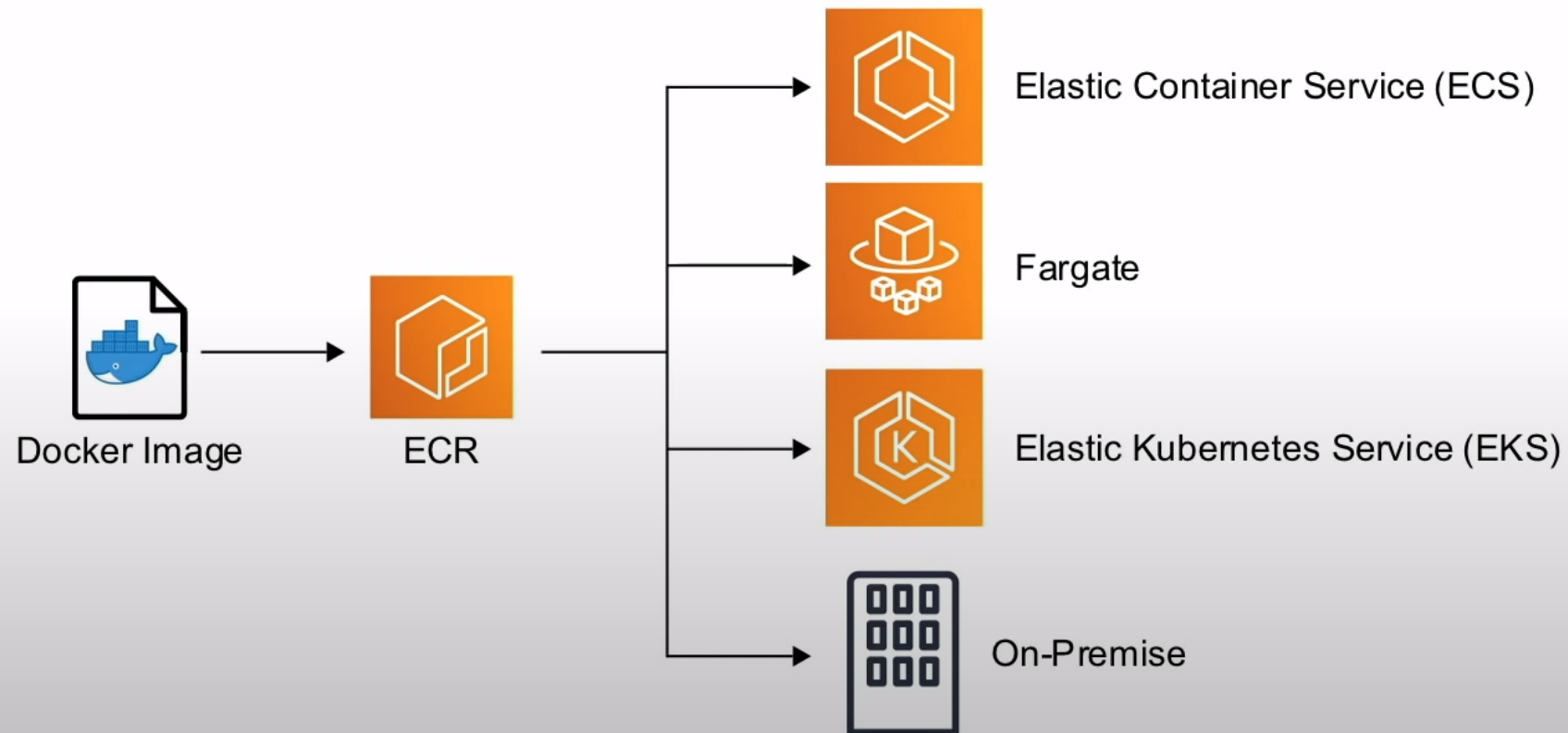


**Elastic Container  
Registry (ECR)**



# Elastic Container Registry (ECR)

A fully-managed Docker container registry that makes it easy for developers to **store**, **manage**, and **deploy** Docker container images.



### Example 1: To create a repository

The following `create-repository` example creates a repository inside the specified namespace in the default registry for an account.

```
aws ecr create-repository \  
  --repository-name project-a/nginx-web-app
```

Output:

```
{  
  "repository": {  
    "registryId": "123456789012",  
    "repositoryName": "sample-repo",  
    "repositoryArn": "arn:aws:ecr:us-west-2:123456789012:repository/project-a/  
nginx-web-app"  
  }  
}
```



# Login To repository

```
aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 201964534042.dkr.ecr.us-east-1.amazonaws.com
```

WARNING! Your password will be stored unencrypted in /home/\*\*\*\*\*/.docker/config.json.

Configure a credential helper to remove this warning. See

<https://docs.docker.com/engine/reference/commandline/login/#credentials-store>

# Create my repository

Follow the command documents:

```
docker tag falcon007/calculator-app:latest 201964534042.dkr.ecr.us-east-1.amazonaws.com/falcon007/calculator-app:latest
docker images
```

| REPOSITORY                                                            | TAG    | IMAGE ID     | CREATED        | SIZE |
|-----------------------------------------------------------------------|--------|--------------|----------------|------|
| 201964534042.dkr.ecr.us-east-1.amazonaws.com/falcon007/calculator-app | latest | 0c9bf8ebd438 | 52 minutes ago | 496M |
| falcon007/calculator-app                                              | latest | 0c9bf8ebd438 | 52 minutes ago | 496M |

```
$ docker push 201964534042.dkr.ecr.us-east-1.amazonaws.com/falcon007/calculator-app:latest
```

```
The push refers to repository [201964534042.dkr.ecr.us-east-1.amazonaws.com/falcon007/calculator-app]
f0178a40a16a: Pushed
56285d9a7760: Pushed
077bff59ce57: Pushed
9cd9df9ffc97: Pushed
latest: digest: sha256:596d44169fd3b41f407468af817059132e228247887f20aaa60e9916aad62253 size: 1166
```

Private | Public

Private repositories (1)



View push commands

Delete

Actions ▼

Create repository

 Find repositories

| <input type="checkbox"/> | Repository name ▲                        | URI                                                                                                                                                     | Created at ▼                      | Tag immutability | Scan frequency | Encryption type | Pull through cache |
|--------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------------------|----------------|-----------------|--------------------|
| <input type="checkbox"/> | <a href="#">falcon007/calculator-app</a> |  201964534042.dkr.ecr.us-east-1.amazonaws.com/falcon007/calculator-app | March 31, 2023, 11:57:40 (UTC+02) | Disabled         | Manual         | AES-256         | Inactive           |



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## Creating an ECS Service



Follow Along

Back

View Cluster

↻ ECS status - 2 of 3 complete **MyECSCluster**



**ECS cluster**

ECS Cluster MyECSCluster successfully created



**ECS Instance IAM Policy**

IAM Policy for the role ecsInstanceRole successfully attached



**CloudFormation Stack**

Creating CloudFormation stack resources

↻ Cluster Resources

Instant  
Desired number of ins  
K  
ECS

# Amazon ECS Instance Role

[PDF](#) | [Kindle](#) | [RSS](#)

AWS Batch compute environments are populated with Amazon ECS container instances, and they run the Amazon ECS container agent locally. The Amazon ECS container agent makes calls to various AWS APIs on your behalf, so container instances that run the agent require an IAM policy and role for these services to know that the agent belongs to you. Before you can create a compute environment and launch container instances into it, you must create an IAM role and an instance profile for those container instances to use when they are launched. This requirement applies to container instances launched with or without the Amazon ECS-optimized AMI provided by Amazon.

The Amazon ECS instance role and instance profile are automatically created for you in the console first-run experience. However, you can use the following procedure to check and see if your account already has the Amazon ECS instance role and instance profile and to attach the managed IAM policy if needed.

## To check for the `ecsInstanceRole` in the IAM console

1. Open the IAM console at <https://console.aws.amazon.com/iam/>.
2. In the navigation pane, choose **Roles**.
3. Search the list of roles for `ecsInstanceRole`. If the role does not exist, use the steps below to create the role.
  - a. Choose **Create Role**.
  - b. For **Select type of trusted entity**, choose **AWS service**. For **Choose the service that will use this role**, choose **Elastic Container Service**. For **Select your use case**, choose **EC2 Role for Elastic Container Service**.
  - c. Choose **Next: Permissions**, **Next: Tags**, and **Next: Review**.
  - d. For **Role Name**, type `ecsInstanceRole` and choose **Create Role**.

Did this page help you?

Yes

No

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Previous topic: [AWS Batch Service IAM Role](#)

Next topic: [Amazon EC2 Spot Fleet Role](#)

Have a question? [Try the forums](#)



**Roles** (5) [Info](#)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.



Delete

Create role

Q

Search

| <input type="checkbox"/> | Role name                                       | Trusted entities                                   | Last activity |
|--------------------------|-------------------------------------------------|----------------------------------------------------|---------------|
| <input type="checkbox"/> | <a href="#">AWSCloud9SSMAccessRole</a>          | AWS Service: cloud9, and 1 more. <a href="#">↗</a> | 2 days ago    |
| <input type="checkbox"/> | <a href="#">AWSServiceRoleForAutoScaling</a>    | AWS Service: autoscaling (Service-Linked Role)     | 2 days ago    |
| <input type="checkbox"/> | <a href="#">AWSServiceRoleForAWSCloud9</a>      | AWS Service: cloud9 (Service-Linked Role)          | 2 days ago    |
| <input type="checkbox"/> | <a href="#">AWSServiceRoleForSupport</a>        | AWS Service: support (Service-Linked Role)         | -             |
| <input type="checkbox"/> | <a href="#">AWSServiceRoleForTrustedAdvisor</a> | AWS Service: trustedadvisor (Service-Linked Role)  | -             |

# Select trusted entity Info

## Trusted entity type

☒ **AWS service**  
Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **AWS account**  
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **Web identity**  
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ **SAML 2.0 federation**  
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**  
Create a custom trust policy to enable others to perform actions in this account.

## Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

### Common use cases

- ☒ **EC2**  
Allows EC2 instances to call AWS services on your behalf.
- ☐ **Lambda**  
Allows Lambda functions to call AWS services on your behalf.

Use cases for other AWS services:

Choose a service to view use case

# Add permissions [Info](#)

## Permissions policies (Selected 1/828) [Info](#)

Choose one or more policies to attach to your new role.



Create policy [↗](#)

Filter policies by property or policy name and press enter.

7 matches

< 1 >

"ec2container"

Clear filters

|                                     | Policy name <a href="#">↗</a>                          | Type     | Description                                                                                                                   |
|-------------------------------------|--------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/>            | <a href="#">AmazonEC2ContainerRegistryFullAccess</a>   | AWS m... | Provides administrative access to Amazon ECR resources                                                                        |
| <input type="checkbox"/>            | <a href="#">AmazonEC2ContainerRegistryReadOnly</a>     | AWS m... | Provides read-only access to Amazon EC2 Container Registry repositories.                                                      |
| <input type="checkbox"/>            | <a href="#">AmazonEC2ContainerServiceEventsRole</a>    | AWS m... | Policy to enable CloudWatch Events for EC2 Container Service                                                                  |
| <input type="checkbox"/>            | <a href="#">AmazonEC2ContainerServiceAutoscaleRole</a> | AWS m... | Policy to enable Task Autoscaling for Amazon EC2 Container Service                                                            |
| <input type="checkbox"/>            | <a href="#">AmazonEC2ContainerRegistryPowerUser</a>    | AWS m... | Provides full access to Amazon EC2 Container Registry repositories, but does not allow repository deletion or policy changes. |
| <input checked="" type="checkbox"/> | <a href="#">AmazonEC2ContainerServiceforEC2Role</a>    | AWS m... | Default policy for the Amazon EC2 Role for Amazon EC2 Container Service.                                                      |
| <input type="checkbox"/>            | <a href="#">AmazonEC2ContainerServiceRole</a>          | AWS m... | Default policy for Amazon ECS service role.                                                                                   |

## ▼ Set permissions boundary - optional [Info](#)

Set a permissions boundary to control the maximum permissions this role can have. This is not a common setting, but you can use it to delegate permission management to others.

- ☒ Create role without a permissions boundary
- ☐ Use a permissions boundary to control the maximum role permissions

Cancel

Previous

Next

# Name, review, and create

## Role details

### Role name

Enter a meaningful name to identify this role.

ecsInstanceRole

Maximum 64 characters. Use alphanumeric and '+=, @- \_ ' characters.

### Description

Add a short explanation for this role.

Allows EC2 instances to call AWS services on your behalf.

Maximum 1000 characters. Use alphanumeric and '+=, @- \_ ' characters.

## Step 1: Select trusted entities

Edit

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "sts:AssumeRole"
8       ],
9       "Principal": {
10        "Service": [
11          "ec2.amazonaws.com"
12        ]
13      }
14    ]
15  }
16 }
```

**Roles (6)** [Info](#)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.



Delete

Create role

 Search

< 1 >



| <input type="checkbox"/> | Role name                                       | Trusted entities                                  | Last activity |
|--------------------------|-------------------------------------------------|---------------------------------------------------|---------------|
| <input type="checkbox"/> | <a href="#">AWSCloud9SSMAccessRole</a>          | AWS Service: cloud9, <a href="#">and 1 more.</a>  | 2 days ago    |
| <input type="checkbox"/> | <a href="#">AWSServiceRoleForAutoScaling</a>    | AWS Service: autoscaling (Service-Linked Role)    | 2 days ago    |
| <input type="checkbox"/> | <a href="#">AWSServiceRoleForAWSCloud9</a>      | AWS Service: cloud9 (Service-Linked Role)         | 2 days ago    |
| <input type="checkbox"/> | <a href="#">AWSServiceRoleForSupport</a>        | AWS Service: support (Service-Linked Role)        | -             |
| <input type="checkbox"/> | <a href="#">AWSServiceRoleForTrustedAdvisor</a> | AWS Service: trustedadvisor (Service-Linked Role) | -             |
| <input type="checkbox"/> | <a href="#">ecsInstanceRole</a>                 | AWS Service: ec2                                  | -             |

**Roles Anywhere** [Info](#)

Manage

# Create cluster [Info](#)

An Amazon ECS cluster groups together tasks, and services, and allows for shared capacity and common configurations. All of your tasks, services, and capacity must belong to a cluster.

## Cluster configuration

Cluster name

There can be a maximum of 255 characters. The valid characters are letters (uppercase and lowercase), numbers, hyphens, and underscores.

## ▼ Networking [Info](#)

By default tasks and services run in the default subnets for your default VPC. To use the non-default VPC, specify the VPC and subnets.

## ▼ Infrastructure [Info](#)

Customized

Your cluster is automatically configured for AWS Fargate (serverless) with two capacity providers. Add Amazon EC2 instances, or external instances using ECS Anywhere.

### ☒ AWS Fargate (serverless)

Pay as you go. Use if you have tiny, batch, or burst workloads or for zero maintenance overhead. The cluster has Fargate and Fargate Spot capacity providers by default.

### ☒ Amazon EC2 Instances

Manual configurations. Use for large workloads with consistent resource demands.

#### Auto Scaling group (ASG) [Info](#)

Use Auto Scaling groups to scale the Amazon EC2 instances in the cluster.

Create new ASG ▼

#### Operating system/Architecture

Choose the Windows operating system or Linux architecture for your instance.

Amazon Linux 2 (arm64) ▼

#### EC2 instance type

Choose based on the workloads you plan to run on this cluster.

t2.micro ▼

#### Desired capacity

Specify the number of instances to launch in your cluster.

Minimum

0

Maximum

1

#### SSH Key pair

Create a key pair in the EC2 console, consisting of a private key and a public key, that you use to prove your identity when connecting to an instance.

None - unable to SSH ▼

### ☐ External instances using ECS Anywhere

Manual configurations. Use to add data center compute.





New ECS Experience

Tell us what you think

Amazon Elastic Container Service

Clusters

Namespaces [New](#)

**Task definitions**

Account settings

Install AWS Copilot

Amazon ECR

Repositories

Amazon Elastic Container Service

Task definitions

Task definitions (0) [Info](#)

Deploy

Create new revision

Create new task definition

Filter task definitions by property or value

No matches

Status of last revision = ACTIVE

Clear filters

| Task definition                                                                                                          | Status of last revision |
|--------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <div><div>No task definitions</div><div>No task definitions to display.</div><div>Create new task definition</div></div> |                         |

Step 1

Configure task definition and containers

Step 2

Configure environment, storage, monitoring, and tags

Step 3

Review and create

# Configure task definition and containers

## Task definition configuration

Task definition family [Info](#)

Specify a unique task definition family name.

calculator

Up to 255 letters (uppercase and lowercase), numbers, hyphens, and underscores are allowed.

## Container - 1 [Info](#)

Essential container

Remove

### Container details

Specify a name, container image, and whether the container should be marked as essential. Each task definition must have at least one essential container.

Name

calculator-app

Image URI

201964534042.dkr.ecr.us-east-1.amazonaws.com/falco

Essential container

Yes



Port mappings [Info](#)

Add port mappings to allow the container to access ports on the host to send or receive traffic. Any changes to port mappings configuration impacts the associated service connect settings.

Host port

0

Container port

80

Protocol

TCP



Port name

calculator-app

App protocol

HTTP



Remove

Add more port mappings



The bridge network mode is selected. By default, dynamic host port mappings are used for each container. To configure a static host port mapping, specify a host port for each container port.

▼ Environment variables - optional [Info](#)

calculator:1

Overview

Info

ARN

arn:aws:ecs:us-east-1:201964534042:task-definition/calculator:1

Task role

-

Status

✔ ACTIVE

Task execution role

[ecsTaskExecutionRole](#) 

Time created

3/31/2023, 10:47:05 UTC

Operating system/Architecture

Linux/X86\_64

App environment

EC2

Network mode

default

Task size

Task CPU

1 vCPU

Task memory

2 GB

Containers

Info

| Container name                 | Image                            | Essential | CPU | Memory | GPU |
|--------------------------------|----------------------------------|-----------|-----|--------|-----|
| <a href="#">calculator-app</a> | 201964534042.dkr.ecr.us-east-... | true      | 0   | -      | -   |

✔ Task definition successfully created

calculator:1 has been successfully created. You can use this task definition to deploy a service or run a task.

Deploy ▾



Amazon Elastic Container Service > Task definitions

Task definitions (1) Info



Deploy ▾

Create new revision ▾

Create new task definition ▾

🔍 Filter task definitions by property or value

1 match

< 1 > ⚙

Status of last revision = ACTIVE 

✕

Clear filters

Task definition ▾

Status of last revision ▾

○ calculator

✔ ACTIVE

calculator:1 has been successfully created. You can use this task definition to deploy a service or run a task.

Amazon Elastic Container Service > Clusters > MyEcs > Services

## MyEcs ASG



Update cluster

Delete cluster

### Cluster overview

|                           |                  |                                  |                                     |
|---------------------------|------------------|----------------------------------|-------------------------------------|
| ARN<br>MyEcs              | Status<br>Active | CloudWatch monitoring<br>Default | Registered container instances<br>- |
| Services<br>Draining<br>- | Active<br>-      | Tasks<br>Pending<br>-            | Running<br>-                        |

Services | Tasks | Infrastructure | Metrics | Scheduled tasks | Tags

### Services (0) [Info](#)



Manage tags

Update

Delete service

Create

Filter services by value

All launch types

All service types



1



|  | Service name | Status | ARN | Service type | Deployments and tasks | Last deploy... | Task defini... | Revision |
|--|--------------|--------|-----|--------------|-----------------------|----------------|----------------|----------|
|--|--------------|--------|-----|--------------|-----------------------|----------------|----------------|----------|

No services

No services to display.

Create

✔ Task definition successfully created  
calculator:1 has been successfully created. You can use this task definition to deploy a service or run a task.

Amazon Elastic Container Service > Clusters > MyEcs > Create service

# Create Info

## Environment

Amazon EC2

### Existing cluster

Select an existing cluster. To create a new cluster, go to [Clusters](#).

MyEcs

### ▼ Compute configuration (advanced)

#### Compute options Info

To ensure task distribution across your compute types, use appropriate compute options.

☒ **Capacity provider strategy**  
Specify a launch strategy to distribute your tasks across one or more capacity providers.

☐ **Launch type**  
Launch tasks directly without the use of a capacity provider strategy.

#### Capacity provider strategy Info

Select either your cluster default capacity provider strategy or select the custom option to configure a different strategy.

- ☒ Use cluster default
- ☐ Use custom (Advanced)

#### Capacity provider

#### Base

#### Weight

Infra-ECS-Cluster-... ▼

0

1

Add more

## Deployment configuration

### Application type [Info](#)

Specify what type of application you want to run.

☐

#### Service

Launch a group of tasks handling a long-running computing work that can be stopped and restarted. For example, a web application.

☒

#### Task

Launch a standalone task that runs and terminates. For example, a batch job.

### Task definition

Select an existing task definition. To create a new task definition, go to [Task definitions](#) .

☐

#### Specify the revision manually

Manually input the revision instead of choosing from the 100 most recent revisions for the selected task definition family.

#### Family

calculator ▼

#### Revision

1 (LATEST) ▼

### Desired tasks

Specify the number of tasks to launch.

1

### Task group

All tasks with the same task group name are considered as a set when performing spread placement.



Create

Info

Environment

Amazon EC2

Existing cluster

Select an existing cluster. To create a new cluster, go to [Clusters](#).

MyEcs

▼ Compute configuration (advanced)

Compute options

Info

To ensure task distribution across your compute types, use appropriate compute options.

☒ Capacity provider strategy

Specify a launch strategy to distribute your tasks across one or more capacity providers.

☐ Launch type

Launch tasks directly without the use of a capacity provider strategy.

Capacity provider strategy

Info

Select either your cluster default capacity provider strategy or select the custom option to configure a different strategy.

- ☒ Use cluster default
- ☐ Use custom (Advanced)

| Capacity provider       | Base | Weight |
|-------------------------|------|--------|
| Infra-ECS-Cluster-... ▼ | 0    | 1      |

Add more



# AWS Developer Associate

ECS Fargate

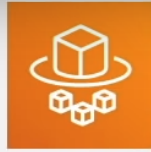


## Introduction to Fargate

# *Fargate*



**Serverless containers.** Don't worry about servers.  
Run containers, and pay based on duration and consumption.



# Introduction to Fargate

- You can create an **empty** ECS cluster (no EC2's provisioned) and then launch Tasks as Fargate
- You **no longer have to provision, configure, and scale** clusters of EC2 instances to run containers
- You are charged for at least one minute and after its by the second.
- You pay based on duration and consumption



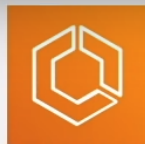


# AWS Developer Associate

ECS Fargate



## Configuring Fargate Tasks



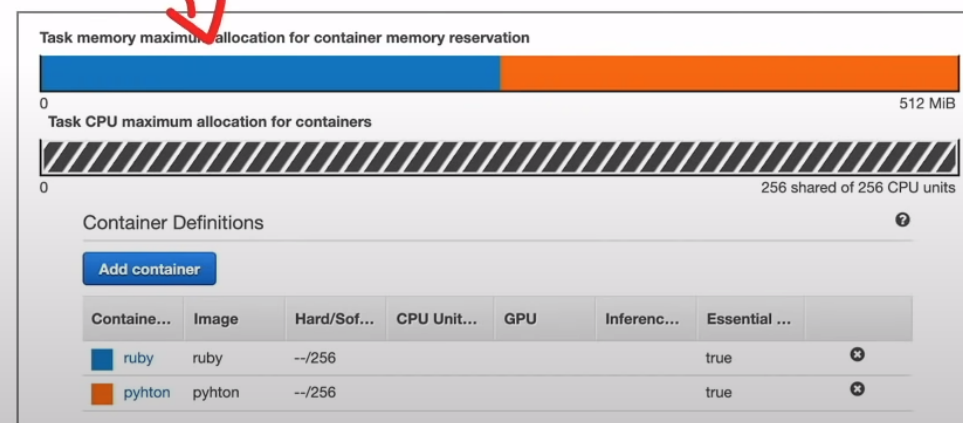
# Configuring Fargate Tasks

- In your Fargate Task Definition you **define** the **memory** and **vCPU**.
- You will then add your containers and **allocate** the memory and vCPU required for each
- When you run the Task you can choose what VPC and subnet it will run in
- **Apply a Security Group to a Task**
- Apply an IAM role to the Task

*\*You can apply SG and IAM role for both ECS and Fargate Tasks and Services*

Task memory (GB) 30GB  
The valid memory range for 4 vCPU is: 8GB - 30GB.

Task CPU (vCPU) 4 vCPU  
The valid CPU for 30 GB memory is: 4 vCPU





# AWS Developer Associate

ECS Fargate



# Fargate Vs Lambda





# Fargate vs



# Lambda

Lambda and Fargate appear very similar, but there are few key differences.

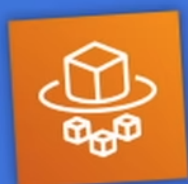
|                    | Fargate                                         | Lambda                                                |
|--------------------|-------------------------------------------------|-------------------------------------------------------|
| <b>Cold Starts</b> | Yes (shorter)                                   | Yes                                                   |
| <b>Duration</b>    | As long as you want                             | <b>15 mins</b> (max)                                  |
| <b>Memory</b>      | Up to <b>30 GB</b>                              | Up to <b>3 GB</b>                                     |
| <b>Containers</b>  | You provide your own containers                 | Limited to standardize containers                     |
| <b>Integration</b> | More manual labour                              | Seamlessly integrates with other serverless services. |
| <b>Pricing</b>     | Pay at least 1 min and every additional second. | Pay per 100ms                                         |





# AWS Developer Associate

ECS Fargate



## Creating a Fargate Service



Follow Along

Clusters > MyFargateCluster > Service: MyFargateService

### Service : MyFargateService

|                  |                      |
|------------------|----------------------|
| Cluster          | MyFargateCluster     |
| Status           | ACTIVE               |
| Task definition  | StudySyncF:1         |
| Service type     | REPLICA              |
| Launch type      | FARGATE              |
| Platform version | LATEST(1.3.0)        |
| Service role     | AWSServiceRoleForECS |

Details Tasks Events Auto Scaling Deployments

#### Load Balancing

| Load Balancer Name | Container Name | Container |
|--------------------|----------------|-----------|
| No load balancers  |                |           |

#### Network Access

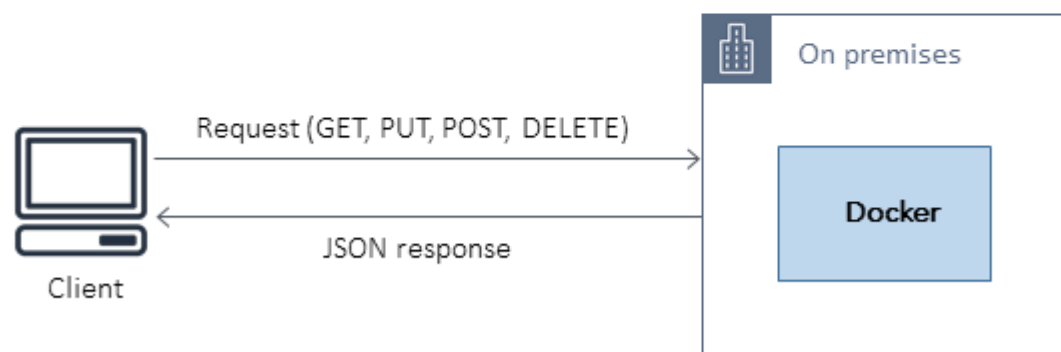
# Deploy Java microservices on Amazon ECS using AWS Fargate

## Architecture

### Source technology stack

- Java microservices (for example, implemented in Spring Boot) and deployed on Docker

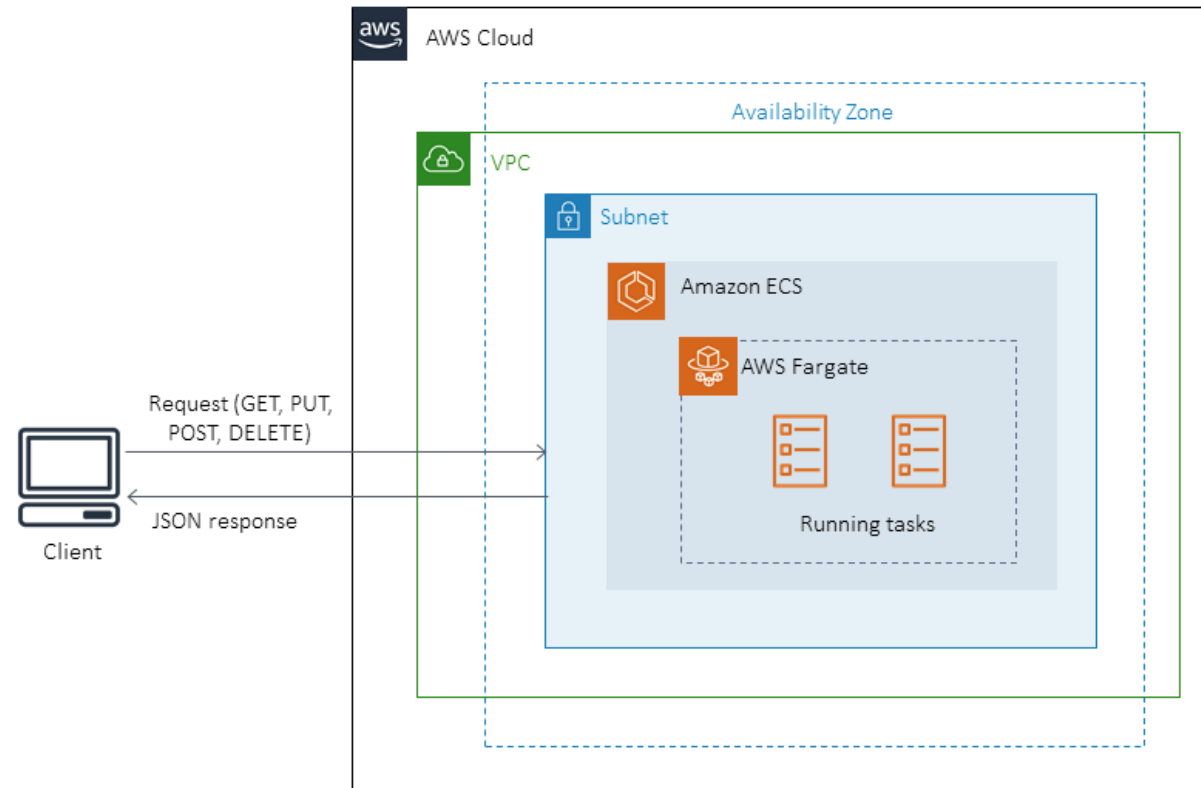
### Source architecture



### Target technology stack

- An Amazon ECS cluster that hosts each microservice by using Fargate
- A VPC network to host the Amazon ECS cluster and associated security groups
- A cluster/task definition for each microservice that spins up containers by using Fargate

### Target architecture



# Tools

## Tools

- [Amazon ECS](#) eliminates the need to install and operate your own container orchestration software, manage and scale a cluster of virtual machines, or schedule containers on those virtual machines.
- [AWS Fargate](#) helps you run containers without needing to manage servers or Amazon Elastic Compute Cloud (Amazon EC2) instances. It's used in conjunction with Amazon Elastic Container Service (Amazon ECS).
- [Docker](#)  is a software platform that allows you to build, test, and deploy applications quickly. Docker packages software into standardized units called *containers* that have everything the software needs to run, including libraries, system tools, code, and runtime.

## Docker code

The following Dockerfile specifies the Java Development Kit (JDK) version that is used, where the Java archive (JAR) file exists, the port number that is exposed, and the entry point for the application.

```
FROM openjdk:11
ADD target/Spring-docker.jar Spring-docker.jar
EXPOSE 8080
ENTRYPOINT ["java", "-jar", "Spring-docker.jar"]
```



# Epics

| ▼ Create new task definitions |                                                                                                                                                                                                                                                                                                                                              |                                          |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Task                          | Description                                                                                                                                                                                                                                                                                                                                  | Skills required                          |
| Create a task definition.     | Running a Docker container in Amazon ECS requires a task definition. Open the Amazon ECS console at <a href="https://console.aws.amazon.com/ecs/">https://console.aws.amazon.com/ecs/</a> , choose <b>Task definitions</b> , and then create a new task definition. For more information, see the <a href="#">Amazon ECS documentation</a> . | AWS systems administrator, App developer |
| Choose launch type.           | Choose <b>Fargate</b> as the launch type.                                                                                                                                                                                                                                                                                                    | AWS systems administrator, App developer |
| Configure the task.           | Define a task name and configure the application with the appropriate amount of task memory and CPU.                                                                                                                                                                                                                                         | AWS systems administrator, App developer |
| Define the container.         | Specify the container name. For the image, enter the Docker site name, the repository name, and the tag name of the Docker image ( <code>docker.io/sample-repo/sample-application:sample-tag-name</code> ). Set memory limits for the application, and set port mappings ( <code>8080, 80</code> ) for the allowed ports.                    | AWS systems administrator, App developer |
| Create the task.              | When the task and container configurations are in place, create the task. For detailed instructions, see the links in the <i>Related resources</i> section.                                                                                                                                                                                  | AWS systems administrator, App developer |

▼ **Configure Task**

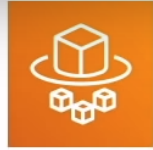
| Task                                                     | Description                                                                                                                                                                                                 | Skills required                          |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Create a task.                                           | Inside the cluster, choose <b>Run new task</b> .                                                                                                                                                            | AWS systems administrator, App developer |
| Choose launch type.                                      | Choose <b>Fargate</b> as the launch type.                                                                                                                                                                   | AWS systems administrator, App developer |
| Choose task definition, revision, and platform version.  | Choose the task that you want to run, the revision of the task definition, and the platform version.                                                                                                        | AWS systems administrator, App developer |
| Select the cluster.                                      | Choose the cluster where you want to run the task from.                                                                                                                                                     | AWS systems administrator, App developer |
| Specify the number of tasks.                             | Configure the number of tasks that should run. If you're launching with two or more tasks, a load balancer is required to distribute the traffic among the tasks.                                           | AWS systems administrator, App developer |
| Specify the task group.                                  | (Optional) Specify a task group name to identify a set of related tasks as a task group.                                                                                                                    | AWS systems administrator, App developer |
| Configure the cluster VPC, subnets, and security groups. | Configure the cluster VPC and the subnets on which you want to deploy the application. Create or update security groups (HTTP, HTTPS, and port 8080) to provide access to inbound and outbound connections. | AWS systems administrator, App developer |
| Configure public IP settings.                            | Enable or disable the public IP, depending on whether you want to use a public IP address for Fargate tasks. The default, recommended option is <b>Enabled</b> .                                            | AWS systems administrator, App developer |
| Review settings and create the task                      | Review your settings, and then choose <b>Run Task</b> .                                                                                                                                                     | AWS systems administrator, App developer |

▼ Cut over

| Task                      | Description                                                                                                               | Skills required                          |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Copy the application URL. | When the task status has been updated to <i>Running</i> , select the task. In the Networking section, copy the public IP. | AWS systems administrator, App developer |
| Test your application.    | In your browser, enter the public IP to test the application.                                                             | AWS systems administrator, App developer |



# ECS and



# Fargate Cheat Sheet

- **Elastic Container Service (ECS)** is fully-managed container orchestration service. Highly secure, reliable, and scalable way to run containers
- Components of ECS
  - **Cluster** Multiple EC2 instances which will house the docker containers.
  - **Task Definition** A JSON file that defines the configuration of (upto 10) containers you want to run
  - **Task** Launches containers defined in Task Definition. Tasks do not remaining running once workload is complete.
  - **Service** Ensures tasks remaining running eg. web-app.
  - **Container Agent** Binary on each EC2 instance which monitors, starts and stops tasks



**Elastic Container Registry (ECR)** A fully-managed Docker container registry that makes it easy for developers to **store**, **manage**, and **deploy** Docker container images.

- **Fargate** is **Serverless containers**. Don't worry about servers. Run containers, and pay based on duration and consumption
  - Fargate has **Cold Starts** so if this is an issue for you then use ECS.
  - **Duration** As long as you want
  - **Memory** Up to 30 GB
  - **Pricing** Pay at least 1 min and every additional second